

SOFIYSK, Russian Federation

WMO: 314780

Lat: 52.2573N

Lon: 133.9960E

Elev: 2961

StdP: 13.19

Time Zone: 10.00 (E10)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-45.0	-42.1	-50.1	0.3	-44.6	-47.0	0.4	-41.9	13.7	8.4	9.8	0.8	2.2	70	0.701

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	78.8	61.4	75.5	60.9	72.1	59.6	64.9	72.7	63.5	71.0	61.9	69.0	4.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.6	94.6	67.7	61.0	89.3	66.2	59.2	83.9	65.1	31.7	71.8	30.5	71.5	29.3	69.2	71.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.3	12.2	9.9	-48.7	84.6	3.5	2.5	-51.3	86.4	-53.3	87.9	-55.3	89.3	-57.8	91.2		
(5)			-48.2	67.7	2.5	1.4	-50.0	68.7	-51.5	69.5	-52.9	70.3	-54.7	71.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.0	-19.0	-12.2	4.3	24.4	43.1	54.7	60.6	57.1	44.3	24.3	-0.4	-19.0		
	DBStd	30.44	9.21	9.57	10.54	9.62	7.09	6.00	4.82	5.81	7.39	9.83	14.17	12.20		
	HDD50	10979	2140	1742	1416	769	234	22	1	12	197	796	1512	2138		
	HDD65	15704	2605	2162	1881	1219	679	313	149	249	621	1261	1962	2603		
	CDD50	770	0	0	0	0	20	162	328	233	27	0	0	0		
	CDD65	19	0	0	0	0	0	3	11	5	0	0	0	0		
	CDH74	434	0	0	0	0	10	111	225	87	1	0	0	0		
	CDH80	54	0	0	0	0	1	10	38	5	0	0	0	0		
(14) Wind	WSAvg	3.7	2.4	3.4	4.0	4.5	5.4	4.0	3.7	3.5	3.9	4.0	3.1	2.5		
(15) Precipitation	PrecAvg	29.1	0.3	0.2	0.4	1.5	2.9	4.2	6.4	6.1	3.9	1.7	0.9	0.4		
	PrecMax	38.3	1.0	0.5	0.9	2.9	5.2	7.6	11.9	13.8	9.2	3.7	2.0	1.1		
	PrecMin	20.4	0.0	0.0	0.0	0.4	1.1	1.5	2.0	1.7	1.1	0.5	0.4	0.0		
	PrecStd	4.4	0.2	0.1	0.2	0.6	1.1	1.7	2.4	2.7	1.6	0.7	0.4	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.4	20.7	37.5	60.2	75.3	81.2	84.5	80.0	70.7	55.5	32.9	20.5		
		MCWB	9.7	17.9	29.7	45.5	54.2	62.9	63.4	63.4	56.7	44.7	30.3	18.7		
	2%	DB	6.5	14.8	31.6	51.3	68.2	77.4	79.8	76.5	65.8	47.1	28.0	14.9		
		MCWB	5.1	12.2	25.4	40.2	50.2	60.2	62.2	61.8	54.5	39.8	25.7	13.2		
	5%	DB	1.5	9.4	27.7	44.9	63.8	73.2	76.0	72.8	61.5	42.2	25.2	8.7		
		MCWB	0.4	7.3	22.8	36.8	47.8	57.3	62.1	60.7	52.3	35.9	23.3	7.4		
	10%	DB	-3.3	5.0	24.0	40.1	58.9	69.3	72.4	69.7	57.9	37.8	21.7	1.4		
		MCWB	-4.1	3.4	19.9	32.7	46.0	56.4	61.4	60.2	50.9	33.1	19.9	0.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.7	18.6	30.3	46.8	55.4	66.7	67.0	66.7	59.3	45.5	31.0	18.5		
		MCDB	11.2	20.4	37.0	58.5	73.3	74.6	74.4	74.5	66.8	52.1	33.2	19.8		
	2%	WB	5.1	12.2	26.1	41.2	52.1	62.6	65.1	64.7	56.4	40.9	26.2	13.4		
		MCDB	6.3	14.9	30.8	50.9	65.2	73.4	72.9	71.7	63.3	46.6	27.8	15.1		
	5%	WB	0.5	7.3	22.9	36.7	49.4	60.1	63.9	63.1	53.9	36.6	23.3	7.4		
		MCDB	1.6	9.2	27.6	44.0	59.9	68.7	71.6	69.1	59.3	41.3	25.0	8.7		
	10%	WB	-4.1	3.4	20.0	32.9	47.2	58.0	62.6	61.2	51.8	33.3	19.8	0.7		
		MCDB	-3.3	5.0	24.0	39.1	56.8	66.4	70.0	67.1	57.0	36.7	21.6	1.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	25.6	29.0	28.8	22.8	21.5	23.4	20.6	21.1	20.5	18.5	21.5	22.8		
		MCDBR	25.5	28.9	29.0	27.8	32.9	32.7	29.8	28.6	28.1	25.0	18.3	18.7		
	5% WB	MCWBR	24.3	26.3	23.3	17.6	16.9	16.1	14.4	15.3	16.9	16.9	16.3	17.7		
		MCWBR	25.3	28.6	27.9	26.0	28.1	25.8	21.4	20.7	23.1	22.5	17.6	19.0		
(40) Clear-Sky Solar Irradiance	taub	taub	0.234	0.266	0.326	0.435	0.488	0.448	0.440	0.379	0.345	0.341	0.275	0.231		
		taud	2.179	2.113	1.961	1.761	1.875	2.119	2.196	2.377	2.404	2.159	2.083	2.125		
	Ebn at Noon	Ebn at Noon	247	268	268	249	243	256	257	269	266	233	217	222		
		Edh at Noon	21	31	45	63	59	47	43	34	29	30	23	19		
(44) All-Sky Solar Radiation	RadAvg	457	828	1345	1655	1561	1697	1562	1345	1018	643	449	349			
	RadStd	12	23	52	126	167	181	129	129	103	54	33	13			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page